

**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES
COMPILER DESIGN (CSA1401)
APPLICATION-BASED QUESTIONS WITH ANSWERS**

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Course: Compiler Design

Course Code: CSA1401

Question 1

Trace the processing of the input expression $P = (a*b) + (b*c)*2$ through all phases of a compiler.

a) Tokens:

$P, =, (, a, *, b,), +, (, b, *, c,), *, 2$

b) Syntax Tree:

Assignment (=)

```

P
+
 *
  a
  b
 *
  *
   b
   c
  2
```

c) Three Address Code:

$t1 = a * b$

$t2 = b * c$

$t3 = t2 * 2$

$t4 = t1 + t3$

$P = t4$

d) Optimized Code:

$t1 = a * b$

$t2 = b * c$

$P = t1 + (t2 * 2)$

e) Significance:

- Produces executable machine code.
- Improves execution speed.
- Optimizes memory usage.
- Generates processor-specific instructions.
- Final output of the compiler.

Question 2

Construct the language and write five valid strings.

a) $(aa)^*$: ϵ , aa, aaaa, aaaaaa, aaaaaaaa

b) $(b^*)abb$: abb, babb, bbabb, bbbabb, bbbbabb

c) $(a^*b^*)^*aba(a^*+b^*)^*$: aba, aaba, abaa, abbaba, aaabbaba

d) $(ab)^*$: ϵ , ab, abab, ababab, abababab

e) $(aba)^* + (a^*b^*)^*$: ϵ , aba, abaaba, aaa, bbbb, aaabbb

Question 3

i) Number of Tokens:

Ignoring comments, the total number of lexical tokens is 30.

ii) Lexical Errors:

The token '1xab' is invalid because identifiers cannot begin with a digit.

The statement $x = 1xab$; therefore contains a lexical error.

Question 4

Regular Expressions:

i) Even number of 0's:

$1^*(01^*01^*)^*$

ii) At least one 1:

$(0+1)^*1(0+1)^*$

iii) Odd number of both 0's and 1's:

$(00)^*0(11)^*1((00)^*(11)^*)^*$

iv) Starts with 1 and ends with 0:

$1(0+1)^*0$

v) No consecutive 1's:

$(0+10)^*1?$

Question 5

Input:

```
int var123 = 12345;
```

```
float var456 = 123.45;
```

```
if(var123 == var456){ return; }
```

a) Total Tokens = 19

b) Token Categories:

Keywords: int, float, if, return

Identifiers: var123, var456

Integer Literal: 12345

Floating Literal: 123.45

Operators: =, =, ==

Separators: (,), {, }, ,, ;

c) FSM State Transitions:

Approximate transitions required = 54.

Reason: Each keyword, identifier, literal, operator and separator requires state transitions for recognition, giving an estimated total of about 54 transitions.